

Data Warehouse SQL Schema

RecoMart Recommendation Pipeline | Task 6 Deliverable | Database: SQLite (data/warehouse.db)

1. Overview

The RecoMart data warehouse is a SQLite database (data/warehouse.db) that stores all engineered features produced by the feature engineering stage. It contains three tables: product_features, user_features, and interaction_matrix. These tables feed directly into the recommendation model training stage.

2. Table: product_features

Source: data/features/product_features.csv | 1,018 rows after deduplication

2.1 Raw Kaggle Columns

Column Name	Type	Description
"Number of clicks on similar products"	REAL	Click engagement signal
"Number of similar products purchased"	REAL	Purchase conversion signal
"Average rating given to similar products"	REAL	Social proof signal
"Gender"	INT	User gender (label-encoded)
"Median purchasing price (in rupees)"	REAL	Price sensitivity signal
"Rating of the product"	REAL	Direct product score (0-5)
"Brand of the product"	INT	Brand identity (label-encoded)
"Customer review sentiment score"	REAL	NLP sentiment (-1 to 1)
"Price of the product"	REAL	Actual price in INR
"Holiday"	INT	0=No, 1=Yes
"Season"	INT	Label-encoded season
"Geographical locations"	INT	Label-encoded region
"Probability for recommendation"	REAL	TARGET variable (0-1)

2.2 Engineered Features

Column Name	Type	Formula / Logic
click_to_purchase_ratio	REAL	clicks / (purchases + 1)
price_bracket	INT	Price quartile: 0=Low, 1=Med, 2=High, 3=Premium
brand_popularity	REAL	Normalised count of products per brand
quality_score	REAL	0.7 x rating + 0.3 x sentiment_score
seasonal_demand	TEXT	"{season}_{holiday}" e.g. "2_1"
rating	REAL	Alias for "Rating of the product"
sentiment_score	REAL	Alias for "Customer review sentiment score"
price	REAL	Alias for "Price of the product"
event_timestamp	TS	Feast compatibility timestamp

3. Table: user_features

Source: data/features/user_features.csv | 200 unique users

Column Name	Type	Formula / Logic
user_id	INT	Primary key - unique user identifier
activity_frequency	INT	COUNT(interactions) per user
avg_rating_given	REAL	MEAN(rating) for rated interactions
rating_variance	REAL	STD(rating) per user - preference consistency
unique_items_interacted	INT	COUNT(DISTINCT product_id) per user
purchase_ratio	REAL	purchase_count / total_interactions
event_timestamp	TS	Feast compatibility timestamp

4. Table: interaction_matrix

Source: data/features/interaction_matrix.csv | 200 users x 1,018 products (sparse)

Column Name	Type	Description
user_id	INT	FK -> user_features.user_id
product_id	INT	FK -> product_features.product_id
rating	REAL	Explicit rating 1.0-5.0 (NULL if interaction was view/click only)

Primary Key: (user_id, product_id) composite

5. Indexes

Index Name	Table	Column	Purpose
idx_interaction_user	interaction_matrix	user_id	Fast user lookup
idx_interaction_product	interaction_matrix	product_id	Fast product lookup
idx_product_rating	product_features	rating	Rating range queries
idx_product_brand	product_features	brand	Brand filter queries

6. Table Relationships

product_features		interaction_matrix		user_features
PK product_id	<	FK product_id		PK user_id
rating		user_id	>	activity_frequency
price		rating		avg_rating_given
quality_score				purchase_ratio
click_to_purchase_ratio				...
...				